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Vision-based hand gesture recognition using deep learning for the interpretation of sign language

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Highlights

- A novel framework for the hand gesture recognition (HGR) using deep learning is presented.
- VGG-11 and VGG-16 has been also modified and implemented for HGR.
- A large dataset of static hand gesture of sign language has been created in this work.
- The remarkable recognition results are obtained by the proposed work over the existing models.

Abstract

Hand gestures have been the key component of communication since the beginning of an era. The hand gestures are the foundation of sign language, which is a visual form of communication. In this paper, a deep learning based convolutional neural network (CNN) model is specifically designed for recognition of gesture-based sign language. This model has a compact representation that

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achieves better classification accuracy with a fewer number of model parameters over the other existing architectures of CNN. In order to evaluate the efficacy of this model, VGG-11 and VGG-16 have also been trained and tested in this work. To evaluate the performance, 2 datasets have been considered. First, in this work, a large collection of Indian sign language (ISL) gestures consisting of 2150 images is collected using RGB camera, and second, a publicly available American sign language (ASL) dataset is used. The highest accuracy of 99.96% and 100% is obtained by the proposed model for ISL and ASL datasets respectively. The performance of the proposed system, VGG-11, and VGG-16 are experimentally evaluated and compared with the existing state-of-art approaches. In addition to accuracy, other efficiency indices have been also used to ascertain the robustness of the proposed work. The findings indicate that the proposed model outperforms the existing techniques as it has the potential to classify maximum gestures with a minimal rate of error. The model is also tested with the augmented data and is found as invariant to rotation and scaling transformation.

Introduction

With the advancement in computer technology and with improvement in hardware equipment, human–computer interaction (HCI) has become a regular part of our lives. In this HCI, the use of hand gestures has gained people's interest as it is a friendly way of interacting with the machine (Just, 2006; Wu et al., 2016). Hand gesture is a way of expressing your idea and thoughts with the use of palm and finger movement (Wu & Huang, 1999). In HCI, one can interact with machines directly through hand movement, without the need for any other input device. For this, a gesture performed by a person needs to be recognized by a machine for meaningful communication between a person and a machine. Hence, hand gesture recognition (HGR) is a hot research topic nowadays and is a need of today’s era. It has many different applications such as virtual Reality (Sagayam & Hemanth, 2017), robot control (Tao et al., 2013), virtual games (Kulshreshth et al., 2017), and natural user interfaces (Ng et al., 2011). A major application of hand gesture recognition is in the human communication system i.e. in sign language recognition (Lichtenauer et al., 2008). Sign language is the visual form of language which uses the structured form of communicative hand gestures to express ideas (Sharma & Singh, 2020). For the speech and hearing impaired people this is the only way of communication. According to the World health organization (WHO), 5% (approx. 360 million) of the total population of the world is suffering from either medium or severe hearing loss and they can communicate only in their respective zonal sign language (World health organization (WHO), 2015). This language is not easily understandable by the hearing community and hence there remains a communication gap between hearing and speech-hearing impaired communities. Thus, gesture recognition using computer technology can be used as an interpreter for sign language translation. This would be advantageous and serve as a link bridge between these communities.

Hand gestures used in sign language can be of two types i.e. static and dynamic gestures. Static gestures are defined as the position of hands and fingers in the space without any movement w.r.t

time, whereas in dynamic gestures there is a continuous movement of hands w.r.t time. The process of recognition of hand gestures for the sign language translation can be done in two different ways namely: vision-based (Moeslund et al., 2006) and contact based (Kumar et al., 2012) recognition. In the contact-based process, signer needs to wear electronic circuitry, for e.g.: data gloves, accelerometer, band, and many such devices. These components measure the change in the movement and transfer the details to the computer for the further processing. This approach has given good recognition result in the literature (Cheok et al., 2019) but it is expensive and inconvenient for the users for daily human–computer interaction interface. The vision-based method is more user-friendly as the signer's data is acquired using a camera. This method reduces the dependency of users on sensory devices and is based on image processing algorithms to process the acquired data. In this paper, a vision-based method has been proposed for the recognition of static hand gestures for sign language translation system. For this, a novel and robust technique based on convolutional neural network (CNN) have been proposed. Along with this proposed method, VGG-11 and VGG-16 have been also modified and implemented for gesture recognition. A broad dataset of static gestures of Indian sign language (ISL) has been collected indigenously in this work for the performance assessment of these approaches, as there is no publicly available dataset of it. This proposed model is found to be advantageous over the present state-of-art approaches since it has high accuracy and consumes less training time. The major contribution and novelty of the paper are as follows:

- A novel and robust model named as G-CNN has been proposed for the hand gesture recognition.
- A dataset consisting of 43 classes of ISL has been collected in this work
- All the hyper-parameters like kernel width, epochs, batch-size, learning-rate are empirically finely-tuned for efficient training of model.
- A thorough experimental analysis of the proposed work has been done by using various evaluation metrics like: accuracy, loss, recognition accuracy of each class, and training time consumed by the model. The results have been also evaluated and compared with VGG-11 and VGG-16 using same dataset.
- Generalization ability of the model has been also proved in this work by evaluating the performance on the augmented dataset. The model shows the competent results and is found to be robust as it is invariant to rotation and scaling transformation.
- The performance of the proposed work has been also evaluated using 10-fold cross validation.
- The work proposed in this model has achieved the remarkable results on each evaluation metrics. Hence it is evident that, this model can easily handle the complexity and hand occlusion of ISL signs.

The remainder of this paper is organized as follows: in Section 2 brief literature review of gesture recognition is presented. In Section 3, the proposed methodology and detail of the collected dataset are presented. Section 4 presents the experimental data and analysis. Finally, the conclusion of this work is given in Section 5.

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Section snippets

Related work

In the human–computer interface, different type of methodologies are available in the literature for the gesture recognition. The primary aim of these methods is to ease the communication by deducing the correct meaning of gestures/signs performed by the user. This process includes the following step namely: acquisition and pre-processing, gesture representation, feature extraction, and classification. This section investigates the various gesture recognition technique in the context of …

Gesture recognition method

The mcethodology for the recognition of static hand gesture is summarized in Fig. 1. In this paper, firstly dataset of static gestures has been collected from multiple signers, and then images are pre-processed in order to accurately extract the gesture detail. Next, images have been fed into the proposed model of CNN for feature learning and the classification has been performed using Softmax classifier. The formation of an appropriate dataset is discussed in Section 3.1. In Section 3.2, the …

Experiment set-up

The proposed work has been implemented on the system with Intel Xeon W-2155 (10 Core, 3.3 GHz), nVIDIA Quadro P4000 8 GB graphics card using Python 3.6 with the Tensorflow and Keras framework. …

Experimental result and analysis

In this paper, two different architecture of CNN (VGG-11 and VGG-16) and proposed model of CNN (G-CNN) has been tested for sign language recognition. The experimental findings for these models are discussed in this section, whereas the comparative study with other state-of-art approaches is presented in …

Conclusion

In this paper, a hand gesture recognition technique is presented for vision-based recognition of sign language. For this, a deep learning based G-CNN model packed with compact representation is proposed. In addition to this, two other architectures VGG11, and VGG-16 have been also examined and modified for the classification of sign language hand gestures. The proposed vision- based model eliminates the user dependency and need of external hardware equipment, thereby making it more practical to …

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CRediT authorship contribution statement

Sakshi Sharma: Conceptualization, Methodology, Software, Writing original draft, Writing review & editing, Visualization, Validation. **Sukhwinder Singh:** Resources, Supervision. …

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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> …CSLR is of special interest to this survey due to its tight connection with the SLT task. Previously, the research interest has been put on the ISLR task and even nowadays it generates some research interest (Cerna, Cardenas, Miranda, Menotti, & Camara-Chavez, 2021; Espejel-Cabrera, Cervantes, García-Lamont, Castilla, & Jalili, 2021; Jenkins & Rashad, 2022; Katılmış & Karakuzu, 2021; Lee, Jo, Kim, Jang, & Park, 2021; Lim, Tan, & Tan, 2016; Neiva & Zanchettin, 2018; Salem, Alharbi, Khezendar, & Alshami, 2019; Sharma & Singh, 2021; Venugopalan & Reghunadhan, 2021; Verma, Aggarwal, & Chandra, 2013). However, the CSLR task has recently received more attention from the research community thanks to the publication of datasets suitable for CSLR such as the RWTH-Phoenix-2014 dataset.…

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